

NATHAN LEBHERZ

Cell & Data Engineer

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PROFESSIONAL EXPERIENCE

Senior Cell Test Engineer

Tesla, Inc

Palo Alto, CA

September 2022 - Present

- Lead engineer for 2 lab locations overseeing 4000 cell test channels for cell abuse, characterization, reliability and aging test methods. Established all equipment bring-up and lab workflows
- Established cell test hardware validation process and created data pipeline with front-end visualization, resulting in 80% time reduction to validate new equipment
- Created Github repository of analysis scripts to analyze, visualize, and validate cell test data
- Designs and writes cell electrical test procedures including fast charge, electrochemical impedance spectroscopy, AC Ripple, thermal runaway and propagation testing
- Developed strategies for testing electrolyte imbalance, post-formation electrolyte removal, and cell short detection in Lithium-Ion Batteries
- Analyzed battery performance, reliability, runaway risk, and CT scan data for presentations to leadership

Software Developer (Freelance Work)

Third Coast Supply Company

Remote

May 2023 – Present

- Built a full-stack Django Application for customers to bid on products listed by Third Coast. The web app allows for mass product uploading, product pictures and is mobile friendly. Over \$1.2 million transacted
- Developed the internal inventory system for the company to track and maintain all products. Functionality allows for barcode printing & scanning, purchase order & invoice generation and financial analytics

Product Engineer

Clarios Power Solutions, LLC

Milwaukee, WI

June 2020 – August 2022

- Member of the College Graduate Rotational Program. The two-year program provided experience covering Product Testing, Chemical Process Development, Product Design, and Materials
- Executed product deviation testing, risk analysis, and reprocessing strategies to mitigate \$1.2M in on-hold product. Created an automated Python and Power BI dashboard for battery formation analysis
- Selected to represent the Company in collaboration with Argonne National Labs to prototype and test cells for research into cycling characteristics. Formally recognized for timely completion of deliverables
- Assembled and tested 2V battery cells by hand-pasting electrode recipes for quick additive comparison
- Deconstructed lead-acid batteries for failure analysis and benchmarking comparisons
- Implemented life-cycling and formation profiles by programming Arbin, MACCOR, and Bitrode cyclers

EDUCATION

Bachelor of Science in Chemical Engineering

Minor in Materials Science & Engineering

University of California, Berkeley

CORE COMPETENCIES

Battery Design & Testing (Li-ion, Abuse, Aging) | Data Analysis & Visualization |
Electrical Test Protocols (EIS, Thermal Runaway, AC Ripple) | Lab Operations & Equipment Validation |
Engineering Leadership & Cross Functional Collaboration | Project Management

TECHNICAL SKILLS

Programming & Tools: Python | MATLAB | SQL | Github | JavaScript | HTML | CSS | SolidWorks | COMSOL
Instrumentation: SEM | NMR | Metallography | Arbin | MACCOR | Bitrode

References Available